

ANCHORAGE ELMENDORF AFB, AK, USA

WMO: 702720

Lat: 61.253N

Lon: 149.794W

Elev: 65

StdP: 100.55

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26401

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.8	-27.7	0.3	-24.7	-25.0	0.4	-21.7	11.4	-0.7	9.2	-2.0	2.3	60	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	23.6	15.0	22.1	14.5	20.2	13.7	16.2	21.2	15.4	19.8	14.7	18.6	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.3	16.5	13.7	9.9	16.0	12.9	9.3	15.6	45.3	21.3	43.2	19.8	41.1	18.7	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.4	6.1	-28.6	26.2	3.6	2.4	-31.1	27.9	-33.2	29.3	-35.2	30.6	-37.8	32.3		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-28.7	17.5	3.5	1.1	-31.2	18.3	-33.2	19.0	-35.2	19.6	-37.7	20.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-9.4	-6.6	-3.5	3.5	9.5	13.9	15.7	14.4	9.6	2.0	-5.6	-8.6
	DBStd	10.25	7.47	6.08	5.22	3.70	2.88	2.52	2.20	2.23	2.89	4.67	5.97	6.44
	HDD10.0	3057	602	464	417	197	43	1	0	1	40	248	467	577
	HDD18.3	5619	861	698	676	446	274	135	86	125	262	506	717	835
	CDD10.0	488	0	0	0	1	28	118	177	136	29	1	0	0
	CDD18.3	9	0	0	0	0	0	2	5	2	0	0	0	0
	CDH23.3	61	0	0	0	0	1	17	29	14	0	0	0	0
	CDH26.7	5	0	0	0	0	0	1	3	1	0	0	0	0
Wind	WSAvg	2.6	2.5	2.5	2.9	2.7	3.0	2.9	2.6	2.3	2.2	2.3	2.4	2.5
Precipitation	PrecAvg	416	20	20	17	13	17	28	46	71	75	49	30	29
	PrecMax	700	69	61	70	48	49	86	114	248	196	109	72	68
	PrecMin	206	0	3	0	0	1	4	5	8	18	9	1	2
	PrecStd	89	14	12	13	12	11	19	25	45	44	24	21	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.3	8.8	16.3	22.2	25.2	26.3	25.1	18.9	12.5	7.8	7.5
		MCWB	2.8	2.6	3.8	8.6	11.8	15.4	16.1	15.4	12.2	7.9	4.3	3.6
	2%	DB	4.1	4.0	6.8	12.8	19.3	22.9	23.6	22.6	17.1	10.5	5.0	3.7
		MCWB	2.2	1.9	2.7	6.3	10.9	14.2	15.5	15.2	11.3	7.2	3.2	1.9
	5%	DB	2.0	2.5	5.0	11.2	17.3	21.1	22.0	20.5	15.5	9.0	3.0	1.6
		MCWB	0.8	1.1	1.6	5.3	9.7	13.3	15.0	14.5	10.9	6.6	1.8	0.6
	10%	DB	0.2	1.1	3.5	9.8	15.3	19.1	20.2	18.8	14.0	7.8	2.0	-0.1
		MCWB	-0.6	0.0	0.9	4.7	8.8	12.5	14.3	13.8	10.4	5.9	1.1	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.4	4.8	9.2	12.7	16.5	17.5	17.2	13.5	9.5	5.0	3.8
		MCDB	5.5	4.9	7.7	15.5	20.2	23.8	23.1	22.5	16.1	11.1	6.6	6.4
	2%	WB	2.1	2.3	3.1	7.1	11.3	15.1	16.4	16.1	12.6	8.1	3.5	2.0
		MCDB	4.0	3.8	5.6	11.6	17.6	21.4	21.5	20.3	15.0	9.3	4.9	3.5
	5%	WB	0.9	1.2	2.2	5.8	10.3	14.1	15.6	15.3	11.9	7.3	2.2	0.6
		MCDB	2.0	2.5	4.5	10.2	16.1	19.4	20.1	18.9	14.3	8.5	3.2	1.4
	10%	WB	-0.6	-0.2	1.2	4.9	9.5	13.1	15.0	14.5	11.2	6.2	1.0	-0.8
		MCDB	0.2	1.0	3.3	9.1	14.6	18.1	18.9	17.7	13.3	7.5	1.7	-0.2
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.3	10.0	9.5	9.5	8.5	7.4	7.7	7.5	6.9	6.8	7.1
		MCDBR	7.5	8.1	9.9	11.7	12.9	12.1	10.7	10.9	9.5	7.7	6.6	7.8
		MCWBR	5.5	6.0	6.4	6.5	6.6	5.5	5.0	5.2	5.2	4.8	4.6	6.0
	5% WB	MCDBR	7.1	7.5	8.5	10.9	11.1	10.4	8.9	8.9	7.7	6.2	6.0	7.0
		MCWBR	5.4	5.6	5.5	6.3	5.9	5.1	4.5	4.8	4.6	4.3	4.4	5.6
Clear-Sky Solar Irradiance	taub	0.231	0.284	0.304	0.345	0.364	0.356	0.368	0.354	0.316	0.295	0.240	N/A	
	taud	2.128	2.238	2.270	2.250	2.317	2.420	2.416	2.463	2.557	2.470	2.124	N/A	
	Ebn at Noon	507	697	815	844	856	872	849	833	813	698	474	N/A	
	Edh at Noon	37	65	88	110	112	103	102	89	68	52	36	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.98	2.59	4.30	5.17	5.32	4.73	3.75	2.44	1.13	0.42	0.15	
	RadStd	0.04	0.13	0.27	0.37	0.46	0.34	0.40	0.40	0.33	0.13	0.06	0.02	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	+2.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+1.98	+1.64	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page